Lecture 4: Neural Networks and Backpropagation

Deep learning models rely on gradient descent to optimize parameters.
Backpropagation is the reverse-mode automatic differentiation algorithm used to compute gradients
Specifically, it applies the chain rule of calculus recursively from the loss layer back to early layers.
The computational graph records forward pass activations for reuse during the backward pass.

Page 2: Optimization and Regularization

Stochastic Gradient Descent (SGD) updates weights using minibatches of data.
Momentum and Adam algorithms accelerate convergence by tracking exponential moving averages
Regularization techniques such as weight decay (L2 penalty) and dropout prevent overfitting in dee
Cross-entropy loss is standard for multi-class classification tasks.